PRACTICE SHEET

1. If the matrix $\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is singular, then what is one of the values of θ?

(a) $\pi/4$ (b) $\pi/2$
(c) π (d) 0

2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}$, If $AB = BA$, then what is the value of x?

(a) -1 (b) 0
(c) 1 (d) Any real number

3. Let $A = (a_{ij})_{n\times n}$ and $adj(A) = (\alpha_{ij})$, If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 2 & 3 & -1 \end{bmatrix}$ then what is the value of α_{23}?

(a) 1 (b) -1
(c) 8 (d) -8

4. Let A and B be two invertible matrices each of order n. What is adj(AB) equal to ?

(a) (adjA) (adjB) (b) (adjA) + (adjB)
(c) (adjA) − (adjB) (d) (adjB) (adjA)

5. M is a matrix with real entries given by

$\begin{bmatrix} 4 & k & 0 \end{bmatrix}$

9. Let A be an m × n matrix. Under which one of the following conditions does A^{-1} exists?

(a) m = n only
(b) m = n and det(A) ≠ 0
(c) m = n and det(A) = 0
(d) m ≠ n

10. Let A and B be two matrices of order n×n. Let A be a non singular. Consider the following:

(a) AB is a singular
(b) AB is a non singular
(c) $A^{-1}B$ is a singular
(d) $A^{-1}B$ is non singular

11. The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ satisfies which one of the following polynomial equations?

(a) $A^2 + 3A + 2I = 0$ (b) $A^2 + 3A - 2I = 0$
(c) $A^2 - 3A - 2I = 0$ (d) $A^2 - 3A + 2I = 0$

12. If a matrix A is symmetric as well as anti-symmetric, then which one of the following is correct?

(a) A is a diagonal matrix
(b) A is a null matrix
(c) A is a unit matrix
(d) A is a triangular matrix

13. If $A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, where ω is cube root of unity, then what is